Docker Networking and Data Handling
IEM 4723 Information Systems Design

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File and Data Handling

Volume Mounting

- When running a container, it is possible to share files with the host.

```
docker run --name container_name --volume \
  /path/on/host/:/path/on/container/ -d image_name:
      latest
```

- This command ensures that the folder or file located at `/path/on/host` is available for use on the container at `/path/on/container`.
- For example

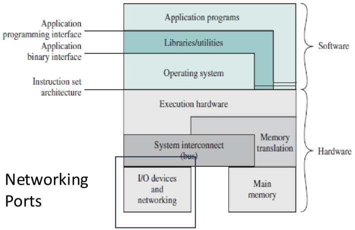
```
docker run --name test_app --volume \
  ~/test_data/:/home/iem5990/ -d test_app:latest
```

Read-Only Volumes

- Volume mounting can also be used for reading data.
- You can specify read and write privileges between the container and the image.
- Use the tag `:ro` to ensure that a container can only read data from the mounted volume, not write to it.

Networking with Docker

Networking Options for Docker



Computing Stack

- It is possible to communicate with other containers, or access services inside a container, through a browser.
- Similar to volumes, ports enable communication across devices.
- You can expose ports on containers to access network services inside and outside the container.
- For example, hosting a website inside the container.

Port Forwarding

- When running a container, it is possible to map ports with the host.

```
docker run --name image_name --publish \
  <host_port>:<container_port> -d image_name:latest
```

- This command ensures that the port `host_port` on the host is available for use on the container at `container_port`.
- For example

```
docker run --name networking_example --publish \
  8201:8501 -d networking:latest
```

Port Forwarding Is Optional

- You can run a container without specifying any port mapping.
- Ports inside the container will then match the host by default.

Today's Examples

You can find these examples at the following link.

`https://github.com/paritoshpr/docker_tutorial`